Rheumatoid Arthritis

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Microbiome Signatures, Microbiome Medicine

Abstract. OverviewRheumatoid arthritis (RA) is a systemic autoimmune disease marked by chronic joint inflammation, synovitis, and bone erosion, driven by Treg/Th17 imbalance, excessive IL-17, TNF-α, and IL-1 production, and macrophage activation. Emerging evidence links microbial dysbiosis and heavy metal exposure to RA, [1][2] with gut microbiota influencing autoimmune activation via Toll-like receptor (TLR) signaling, inflammasome activation, [...]

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Overview

Rheumatoid arthritis (RA) is a systemic autoimmune disease marked by chronic joint inflammation, synovitis, and bone erosion, driven by Treg/Th17 imbalance, excessive IL-17, TNF-α, and IL-1 production, and macrophage activation. Emerging evidence links microbial dysbiosis and heavy metal exposure to RA, [1][2] with gut microbiota influencing autoimmune activation via Toll-like receptor (TLR) signaling, inflammasome activation, and molecular mimicry, contributing to systemic immune dysfunction.

Associated Conditions

Conditions such as periodontitis, autoimmune thyroid disorders (Hashimoto's, Graves'), multiple sclerosis (MS), and inflammatory bowel disease (IBD). RA is also linked to nonalcoholic fatty liver disease (NAFLD), fibromyalgia, and chronic fatigue syndrome (CFS). These associations highlight RA's systemic complexity beyond joint inflammation.

Causes

Rheumatoid arthritis (RA) is a complex, multifactorial disease influenced by genetic predisposition, immune dysregulation, environmental exposures, and metabolic factors. While traditional theories emphasize the role of HLA-DR4 and HLA-DR1 genetic risk factors, along with autoantibody production and inflammatory cytokines such as TNF-α, IL-6, and IL-1β, a more holistic approach integrates emerging insights into microbial-metal interactions and their contribution to disease progression. Heavy metal dysregulation, in particular, has been implicated in immune dysfunction, oxidative stress, and the disruption of gut and oral microbiota, which collectively exacerbate RA pathogenesis.

Primer

Microbial-metal interactions play a significant role in the pathogenesis of rheumatoid arthritis (RA), with heavy metal dysregulation contributing to immune dysregulation and oxidative stress. High soil copper

(Cu) exposure has been correlated with increased inflammation in RA patients, with Cu levels positively associated with erythrocyte sedimentation rate (ESR), platelet count, and disease activity (DAS28). [3]Lead (Pb) and cadmium (Cd) further exacerbate immune dysfunction, with Cd inhibiting superoxide dismutase, thereby reducing antioxidant defenses and increasing joint damage. [4][5] Elevated nickel (Ni) levels in RA patients have been linked to inflammasome activation, while increased chromium (Cr) concentrations affect immune tolerance and oxidative stress. [6] Additionally, microbial dysbiosis influences metal homeostasis, as *Porphyromonas gingivalis* enhances metal-driven citrullination, accelerating autoimmunity. [7] Gut microbial shifts, particularly *Prevotella* overgrowth and *Akkermansia* depletion, further impact metal absorption and immune function, reinforcing the intricate relationship between heavy metal exposure, microbial imbalances, and RA pathogenesis. [8]

Metal Homeostasis in RA

Metal	Mechanism
Copper (Cu)	Promotes IL-6, TNF-α, neutrophil activation. [9]
Lead (Pb)	Induces ROS, disrupts immune tolerance. [10]
Cadmium (Cd)	Inhibits SOD, increases oxidative stress. [11]
Nickel (Ni)	Induces IL-1β, Th17 responses. [13]
Chromium (Cr)	Alters mitochondrial function, immune suppression. [14]

Interventions

Microbiome-targeted interventions offer a promising avenue for modulating immune responses in rheumatoid arthritis (RA), with probiotic therapy emerging as a potential intervention. *Parabacteroides histicola* has been shown to restore regulatory T cell (Treg) function, counteracting the pro-inflammatory effects of T helper 17 (Th17) cells, which play a central role in RA pathogenesis. By promoting immune tolerance and reducing systemic inflammation, *P. histicola* may help mitigate disease progression, offering a microbiome-based approach to complement conventional treatments.

Update History

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Rheumatoid Arthritis major
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Hashimoto's Thyroiditis
Hashimoto's Thyroiditis (HT) is an autoimmune disease that progressively damages the thyroid, often causing hypothyroidism and affecting women disproportionately. Research links HT to gut dysbiosis via the

gut–thyroid axis and highlights heavy metals like nickel, arsenic, and lead as contributors to oxidative stress and thyroid dysfunction.

Lead (Pb)

Lead exposure has a profound effect on the microbiome, disrupting microbial diversity, immune responses, and contributing to the development of antimicrobial resistance (AMR). Understanding how Pb interacts with microbial communities and impacts host-pathogen dynamics is essential for clinicians to mitigate long-term health risks and improve treatment strategies.

Cadmium (Cd)

Cadmium (Cd) is a highly toxic heavy metal commonly found in industrial, agricultural, and environmental settings. Exposure to cadmium can occur through contaminated water, food, soil, and air, and it has been linked to a variety of health issues, including kidney damage, osteoporosis, and cancer. In agriculture, cadmium is often present in phosphate fertilizers and can accumulate in plants, entering the food chain. Its toxicity to living organisms makes cadmium a subject of regulatory concern worldwide, particularly in industrial waste disposal and environmental monitoring.

Nickel

Bacteria regulate transition metal levels through complex mechanisms to ensure survival and adaptability, influencing both their physiology and the development of antimicrobial strategies.

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A microbiome signature is defined through a comprehensive approach that integrates differential abundance analysis across multiple datasets, followed by interpretation for biological relevance. This rigorous process ensures that the identified signature is both statistically robust and biologically

meaningful. Associations that do not meet statistical significance or lack mechanistic relevance are excluded from the final signature.

Relative Abundance (Class)

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Relative Abundance (Family)

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Relative Abundance (Genus)

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Relative Abundance (Kingdom)

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Relative Abundance (Order)

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Relative Abundance (Phylum)

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Relative Abundance (Species)

▶ **Studies**

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